

Topic 6: Problem solving with programming

All problems set in the practical programming tasks in the examination can be solved with the functionalities presented in the Programming Language Subset (PLS) document provided on the [GCSE in Computer Science section of our website](#).

Subject content	Students should:
6.1 Develop code	6.1.1 be able to use decomposition and abstraction to analyse, understand and solve problems
	6.1.2 be able to read, write, analyse and refine programs written in a high-level programming language
	6.1.3 be able to convert algorithms (flowcharts, pseudocode*) into programs
	6.1.4 be able to use techniques (layout, indentation, comments, meaningful identifiers, white space) to make programs easier to read, understand and maintain
	6.1.5 be able to identify, locate and correct program errors (logic, syntax, runtime)
	6.1.6 be able to use logical reasoning and test data to evaluate a program's fitness for purpose and efficiency (number of compares, number of passes through a loop, use of memory)

*In this specification, the term 'pseudocode' is used to denote an informal written description of a program. Pseudocode uses imprecise English language statements and does not require any strict programming syntax.